



GCP Vertex AI Pipelines

Google Cloud's Managed MLOps Orchestration

A comprehensive solution for streamlining your ML workflows from development to production at scale

Why Vertex AI Pipelines?

Challenge

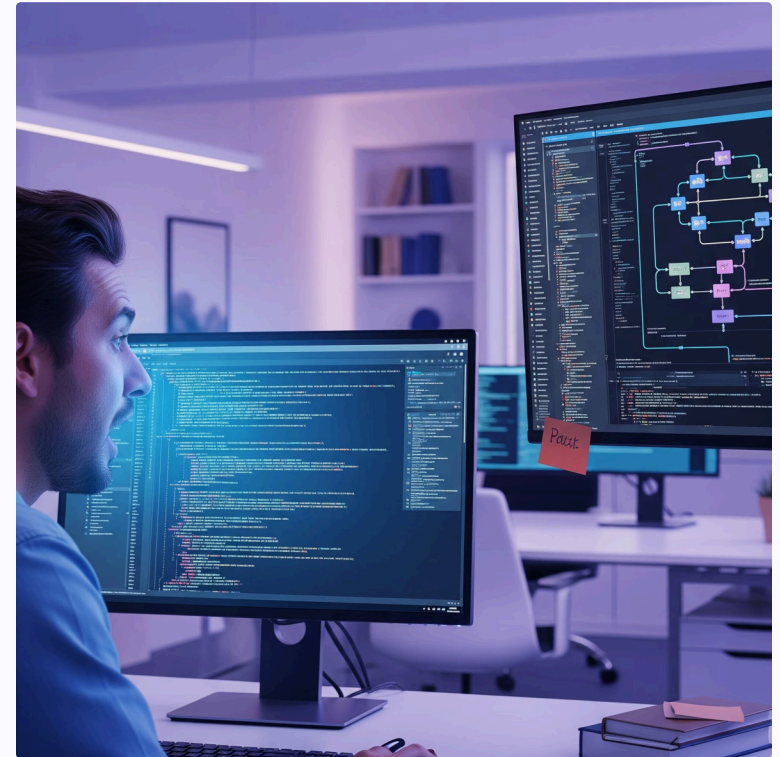
ML lifecycle requires **scalable automation** across multiple stages

Ad-hoc scripts create technical debt and lack reproducibility

Solution

Vertex AI Pipelines = **fully managed Kubeflow Pipelines**

Seamlessly integrated with BigQuery, GCS, AutoML, and GCP infrastructure



Core Capabilities



Pipeline Definition

Define ML workflows as **directed acyclic graphs (DAGs)** with clear dependencies between steps



Metadata Storage

Automatically track artifacts, lineage, and metrics for every pipeline run to ensure reproducibility



Orchestration

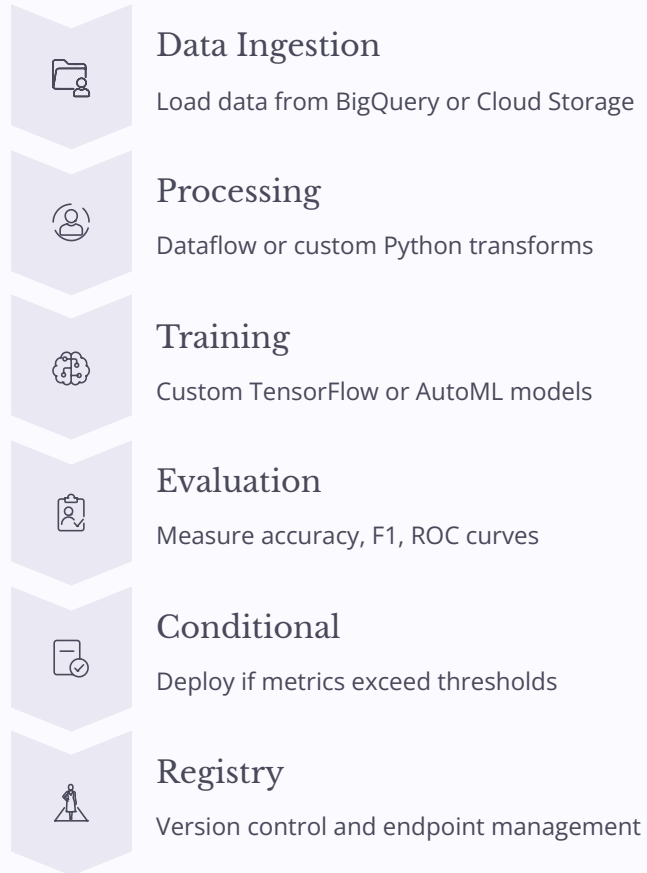
Seamlessly coordinate preprocessing, training, evaluation, and deployment steps across Google Cloud services



Serverless Infrastructure

Run on **fully managed infrastructure** with no Kubernetes expertise required

Pipeline Workflow Example



How It Works

Define in Python

Create pipelines using the [KFP SDK](#) with standard Python code

Compile to YAML

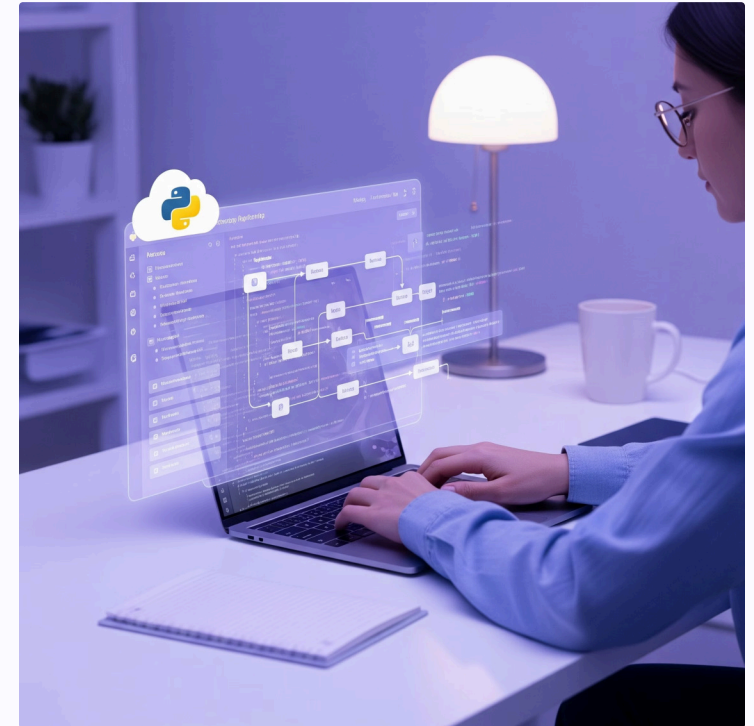
Transform Python definitions into YAML specifications for execution

Vertex AI Orchestration

Managed service schedules and executes jobs across GCP services

Monitor via UI

Visualize pipelines, track lineage, and compare runs through the console



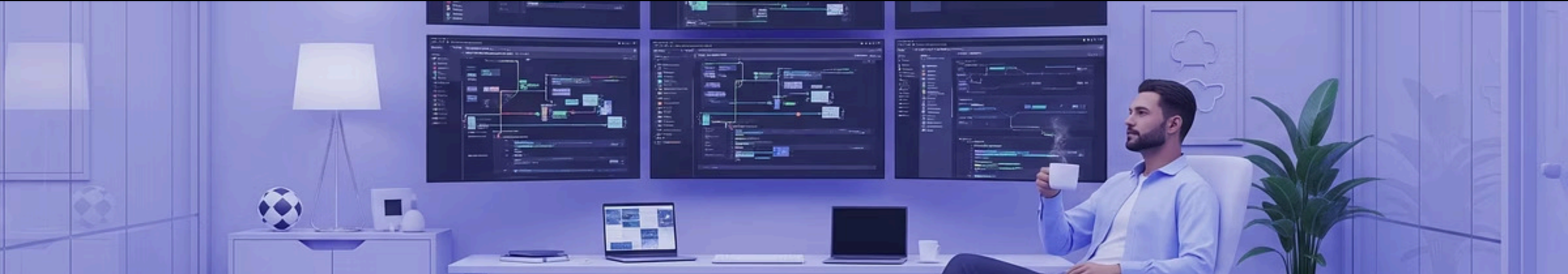
No infrastructure management required - focus on your ML logic

Example Code Snippet

```
from kfp.v2 import dsl
from google_cloud_pipeline_components import aiplatform as gcc_aip

@dsl.pipeline(name="vertex-pipeline")
def pipeline(project: str, location: str):
    train_task = gcc_aip.CustomTrainingJobRunOp(
        display_name="train-model",
        project=project,
        location=location,
        model_display_name="my-model",
        worker_pool_specs=[{
            "machine_spec": {"machine_type": "n1-standard-4"},
            "replica_count": 1,
            "python_package_spec": {
                "executor_image_uri": "gcr.io/my-image",
                "package_uris": ["gs://my-bucket/train.tar.gz"],
                "python_module": "train"
            }
        }]
    )
```

This code defines a simple training pipeline that runs a custom model training job on an n1-standard-4 machine. The **CustomTrainingJobRunOp** component handles infrastructure provisioning, job execution, and cleanup automatically.



Key Benefits

Serverless

No Kubernetes cluster management or infrastructure overhead

Scalable

Effortlessly scale from single CPU jobs to massive TPU pod training

Integrated

Native connections to BigQuery, Dataflow, Looker, and Cloud Storage

Governed

Built-in model registry, lineage tracking, and comprehensive audit logs

Observability & Monitoring

Vertex ML Metadata

Automatically tracks inputs, outputs, and parameters for every pipeline run

Evaluation Metrics

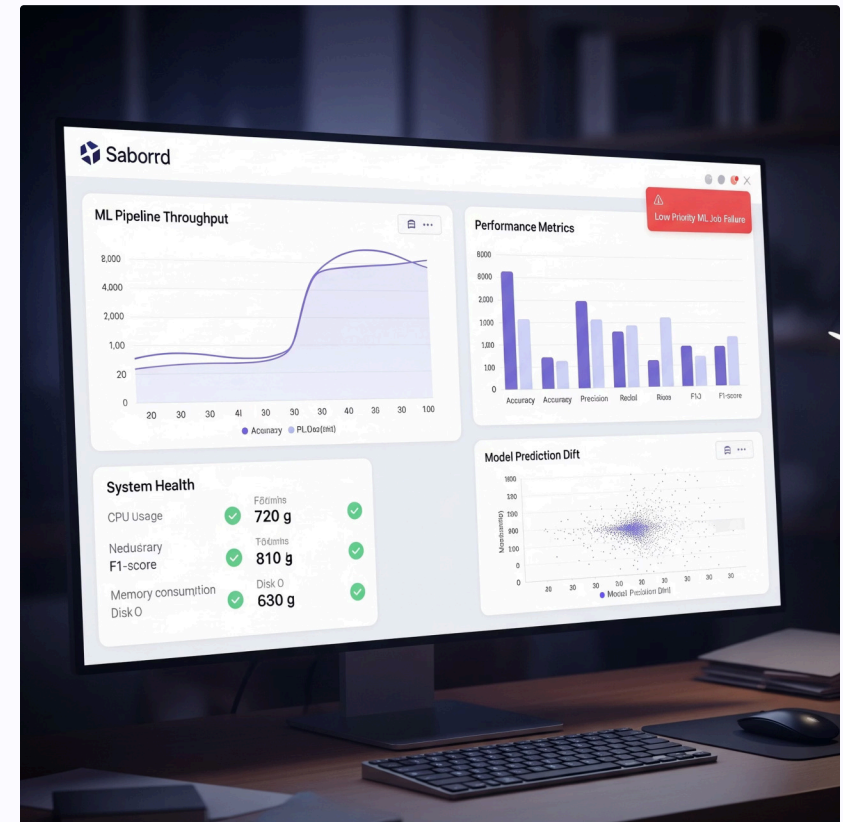
Model performance metrics logged directly to ML Metadata for comparison

Cloud Monitoring Integration

Track infrastructure health, latency, and cost with built-in dashboards

Alerting System

Configure notifications for pipeline failures or model drift detection



End-to-end visibility into your ML workflows

Common Use Cases



Automated Retraining

Schedule pipelines to retrain models whenever new data arrives in BigQuery, ensuring models stay current



Hybrid Model Approach

Combine [AutoML models](#) with custom TensorFlow components in the same workflow



Explainable & Auditable ML

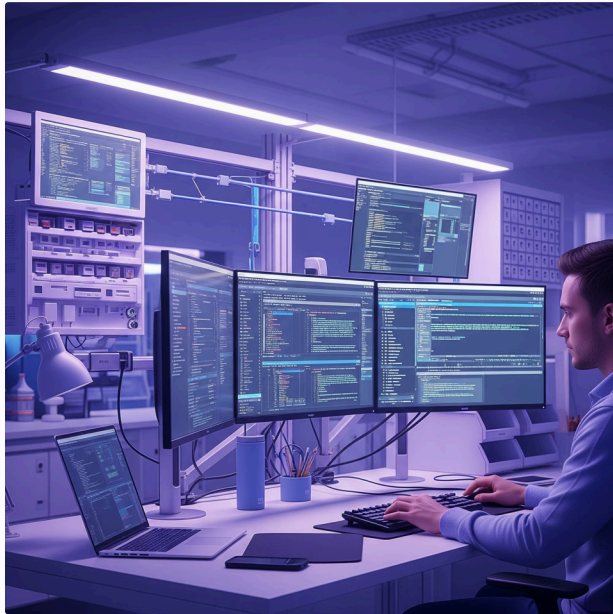
Build systems with integrated explainability tools and comprehensive audit trails



Research to Production

Scale academic ML research to enterprise-grade production systems on GCP

Best Practices



Parameterize Everything

Use **pipeline parameters** for training configs, dataset paths, and hyperparameters to maximize reusability

Version Control Data

Store all datasets in **GCS buckets with versioning** enabled to ensure reproducibility

Log Everything

Always log evaluation metrics, model parameters, and dataset statistics into ML Metadata

Approval Workflows

Use Vertex AI Model Registry to implement human-in-the-loop approval before production deployment



Key Takeaways



Managed MLOps

Vertex AI Pipelines provides **managed Kubeflow Pipelines** on GCP without infrastructure overhead



End-to-End Automation

Seamlessly automate the entire ML lifecycle from data preparation to deployment to monitoring



Deep Integration

Built for tight integration with **BigQuery, GCS, AutoML** and the entire Google Cloud ecosystem



Enterprise-Ready

Designed for **GCP-first enterprise ML operations** with governance, scale, and security built-in